The Effect of Proteasome Inhibitors on the HSF1 Stress Response of the Sea Anemone *Nematostella vectensis* when Introduced to Heat Shock

Introduction

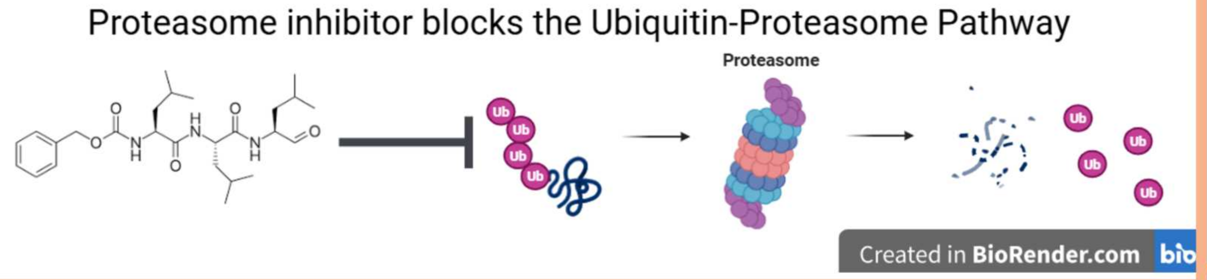
- Coral Reefs are necesary for marine diversity as they support over 25% of marine life. Sadly, global warming has caused ocean temperatures to rise, which causes coral bleaching
- Coral is unable to regenerate in stressful conditions, which leads to its death
- An estimated 30% of coral reefs have died (NOAA, 2025)

NOAA, 2025)

Background

- **Heat Shock Factors** (HSFs) are genes that regulate expression of Heat Shock Proteins (HSPs). HSPs regulate the cellular response to environmental stress
- **HSF1** is a gene in charge of responding to **thermal stress**.
- The **ubiquitin-proteasome pathway** is a system where proteins are broken down by proteasome molecules. **Proteasome inhibitors** block the ubiquitin-proteasome pathway, which increases protein expression, as less proteins are broken down.
- **MG-132** is peptide aldehyde protein inhibitor commonly used in experiments
- *Nematostella vectensis* is a sea anemone commonly used as a model for coral

Proteasome inhibitor blocks the Ubiquitin-Proteasome Pathway



Purpose

- If proteasome degredtion inhibitors are added, then the expression of HSF1 will increase and the stress of the *N. vectensis* will lower. This is because protein degradation inhibitors block the ubiquitin pathway of breaking down proteins, which should result in an overexpression of a targeted gene. HSF1 is in charge of thermal stress, so an overexpression should increase its tolerance to warmer temperatures
- The purpose of this project is to show that increasing tolerance to heat reduces stress sensitivity and to apply this concept to coral-bleaching prevention efforts

Variables

Independent variable
- The presence of MG-132

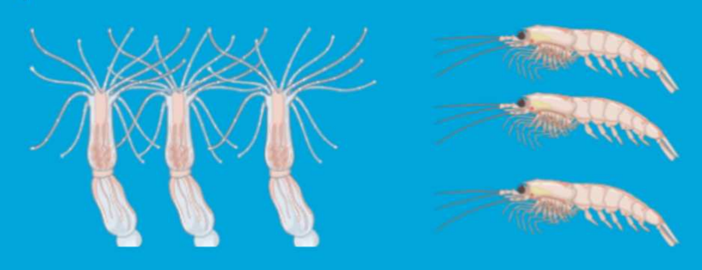
Dependent variable
- Gene expression of HSF1 by RT-qPCR

Controls
- Positive control
  - No MG-132 and no induced heat shock
- Negative control
  - No MG-132 and induced heat shock
- qPCR: NvActin is a housekeeping gene (a control)

Methods

1 Culture

- Nematostella were cultured and seperated into containers for control groups 1 and 2, and experimental

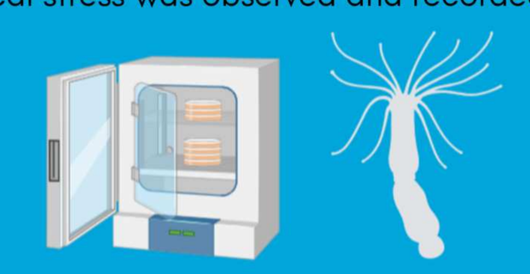


2 Toxicity Assay

- A toxicity assay was performed for MG-132 on Nematostella.
- Concentrations of 0.1,0.5,1,5,10 μM were used

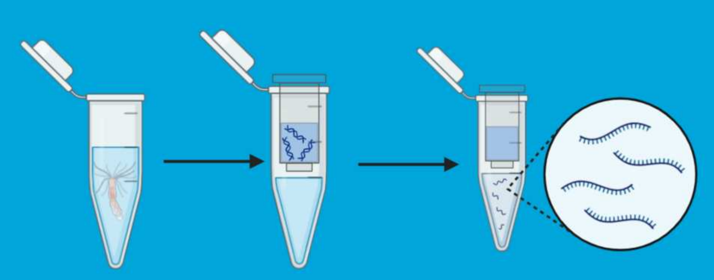
3 Induce Stress

- The control 2 and experimental groups were exposed to heat shock between 35-40 degrees celsius.
- Physical stress was observed and recorded



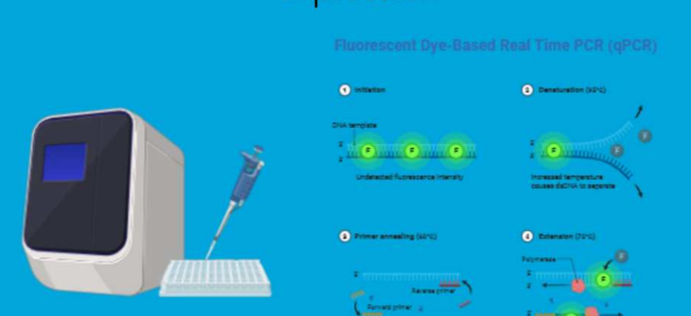
RNA Extraction 4

- 3 Nematostella from each trial and group were removed and RNA was extracted
- RNA concentration was measured using a qubit fluorometer



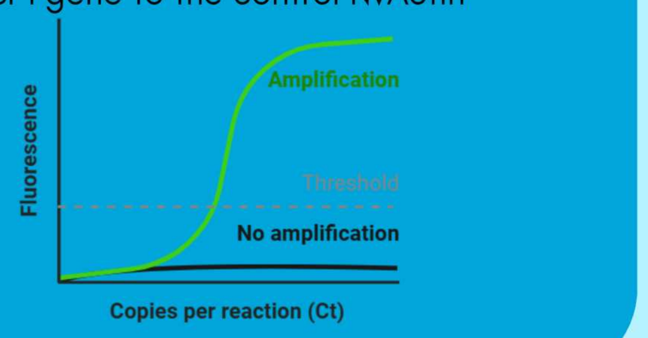
RT-qPCR 5

- Primers were added to RNA samples
- RNA was reverse transcribed into cDNA and amplified by RT-qPCR to measure gene expression



Analysis 6

- Amplification plot were produced to visualize the gene expression
- ΔΔCt Analysis was perfromed to compare the target HSF1 gene to the control NvActin



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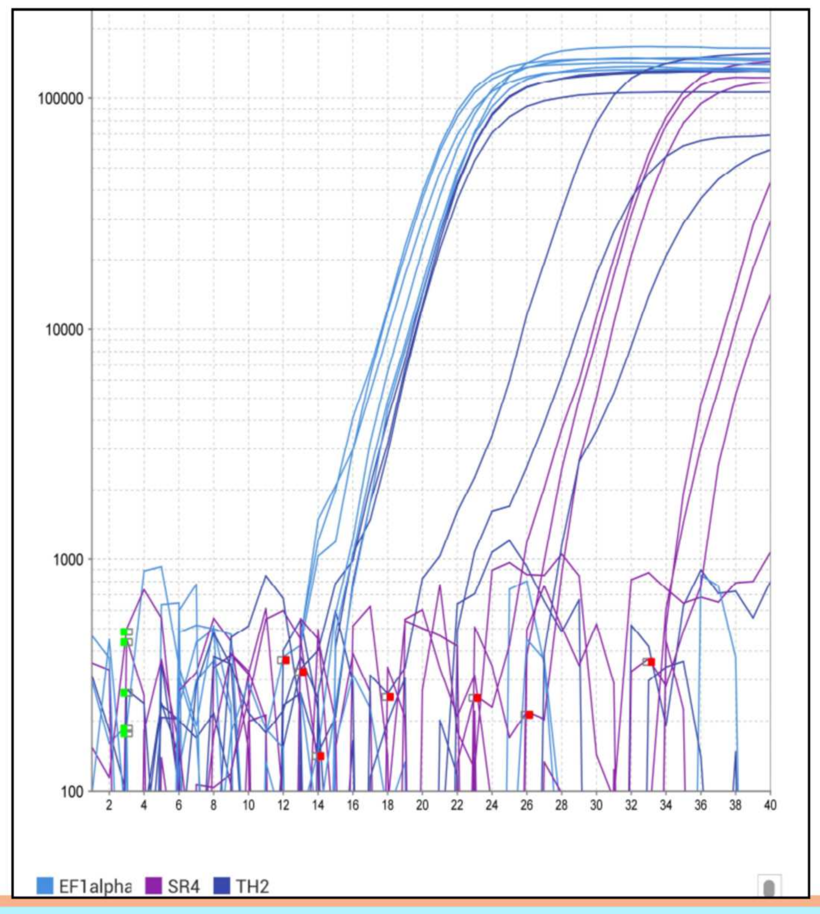
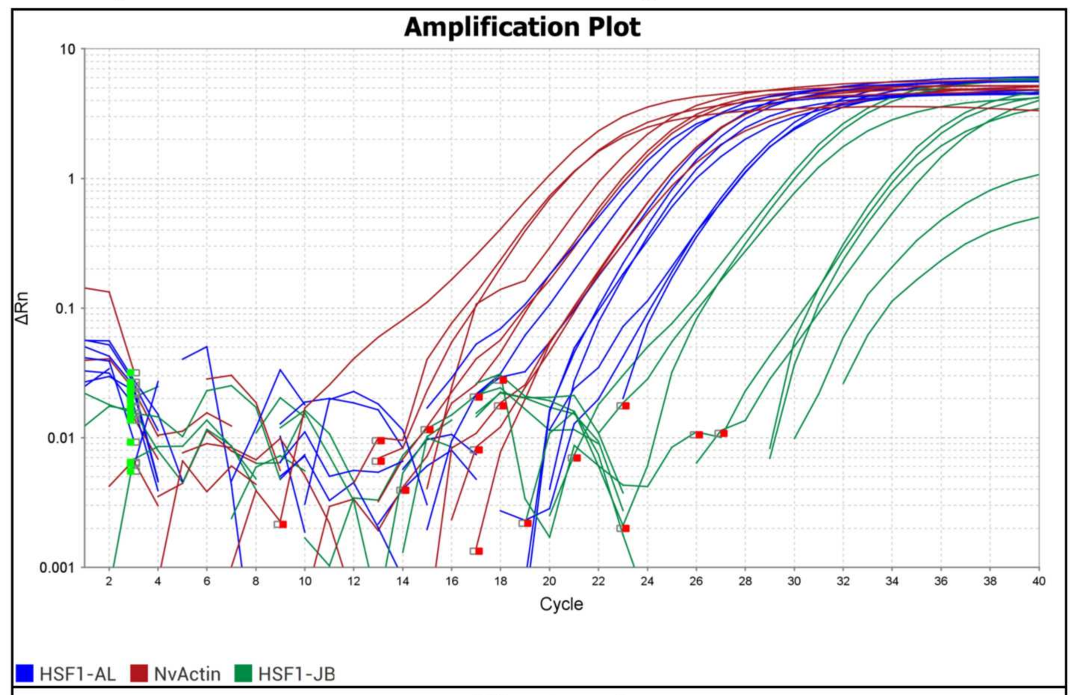
Results

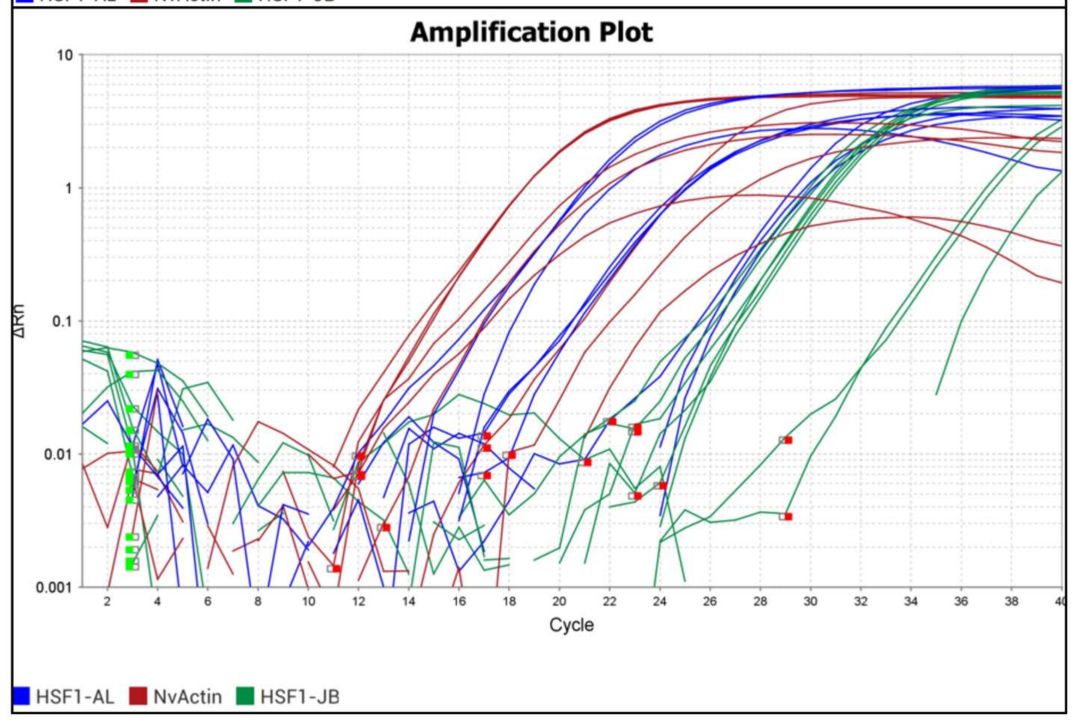
Amplification Plots for qPCR of Control 1, 2, experimental of HSF1 and control genes

Unstressed Nematostella

Stressed Nematostella







Conclusion

There were multiple errors with the qPCR that were removed to give more accurate results. Ct values over 35 or with many errors were removed and the ΔΔCt analysis was run. Overall, the results of the qPCR test and ΔΔCt analysis loosely support the initial hypothesis.

Trial 1

- **Exp vs. C1**
  - Upregulated 1.4 x relative to the control
- **Exp vs. C2**
  - Upregulated 13 x relative to the control
- **C2 vs. C1**
  - Downregulated 10% from the control

Trial 2

- **Exp vs. C1**
  - Upregulated 12.21 x relative to the control
- **Exp vs. C2**
  - Upregulated 2.46 x relative to the control
- **C2 vs. C1**
  - Upregulated 4.95 x relative to the control

There was significant variance between the upregulated genes in trials 1 and 2 of the qPCR. However, both trials resulted in an upregulation of the experimental group compared to both the control 1 and control 2 groups. This indicates that the MG-132 caused HSF1 to be more present both when exposed and not exposed to heat shock. This means that there should be an increased response to thermal stress, which would make the thermal stress decrease. While there were errors, the initial hypothesis was loosely supported as HSF1 expression increased

Additional analysis was run on the physical stress observations of the *N. vectensis* using the tentacle count. Although Control 1 was expected to more closely resemble the experimental group, statistical analysis with a kruskal-wallis test showed that the experimental group differed significantly from both controls ($p < 0.001$, $p=0.04$), while the two control groups, C1 and C2, were not significantly different from each other ($p=0.59$). This did not support the hypothesis because the control 2 was also expected to be different from the control 1. A larger sample size and longer trial duration would be needed to further determine if the thermal stress lowered with the increased HSF1 expression

Limitations

- Errors in the qPCR likely resulted from affected primer samples and a small sample size
- The outlier errors that significantly affected the results were removed so that data was able to be properly interpreted
- Future work can be performed to increaes the sample size in order to make the results more accurate and precise.

Future Work

- This project is intended to propose a solution to increase coral tolernace to heat shock. This has potential to reduce coral death in bleaching events
- Future work can be performed to determine the effects MG-132 has on the regeneration rates while exposed to heat shock

References

Lee, D. H., & Goldberg, A. (1998, October 01). Proteasome inhibitors: valuable new tools for cell biologists. Trends in Cell Biology, 8(10), 397-403. 10.1016/S0962-8924(98)01346-4

Pirkkala, L., Alastalo, T.-P., Zuo, X., Benjamin, I., & Sistonen, L. (2000, January 13). Disruption of Heat Shock Factor 1 Reveals an Essential Role in the Ubiquitin Proteolytic Pathway. Molecular and Cellular Biology, 20(8). 10.1128/mcb.20.8.2670-2675.2000

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